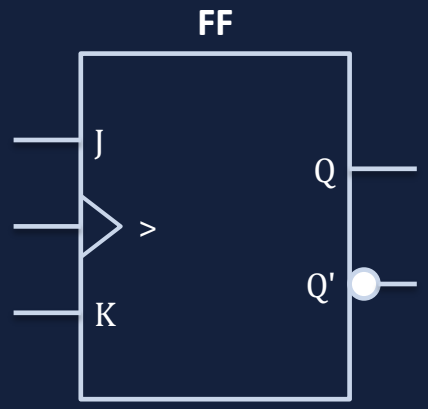
ENES 246 · DIGITAL LOGIC DESIGN

Lecture 23

Review for Exam 3



Latches and flip-flops · timing · sequential analysis · state reduction and design · registers and counters · practice set

Reading: Mano & Ciletti, *Digital Design with an Introduction to the Verilog HDL*, 6th ed. — Chapters 5–6 (review)

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Today’s roadmap

1 §5.4

Storage elements

Lectures 18–19

2 §5.5

Analysis

Lecture 19

3 §5.8

Reduction, design

Lecture 20

4 §6.4

Counters

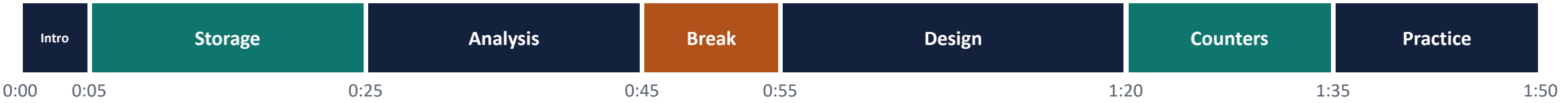
Registers, counters: 21–22

By the end of class you can…

- Draw latch and flip-flop outputs from a timing diagram
- Compute maximum clock frequency and check hold
- Analyze a clocked circuit to a state table and diagram
- Reduce states and design with D, JK or T flip-flops
- Trace shift registers and design counters

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Agenda for today’s session



Elapsed time	Topic	In-class work
0:00 – 0:05	Welcome; what the exam covers	—
0:05 – 0:25	Latches, flip-flops, timing	Reviews 1–3
0:25 – 0:45	Sequential analysis	Review 4
0:45 – 0:55	Break	
0:55 – 1:20	State reduction and design	Reviews 5, 6
1:20 – 1:35	Registers and counters	Reviews 7, 8
1:35 – 1:50	Timed practice set and solutions	Practice set

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What Exam 3 covers

Lecture	Topic	You should be able to
18	Latches	trace SR, S'R' and D latches; transparency
19	Flip-flops, analysis	characteristic equations; state tables and diagrams; Mealy vs Moore
19	Timing	setup, hold, f_{max}
20	Reduction, design	implication table; D, JK and T design; unused states
21	Registers	parallel load; shift registers; serial adder; ripple counters
22	Counters	synchronous, up-down, BCD, arbitrary, ring, Johnson

Memorize: $Q^+ = D$, $Q^+ = JQ' + K'Q$, $Q^+ = T \oplus Q$ and the JK and T excitation tables.

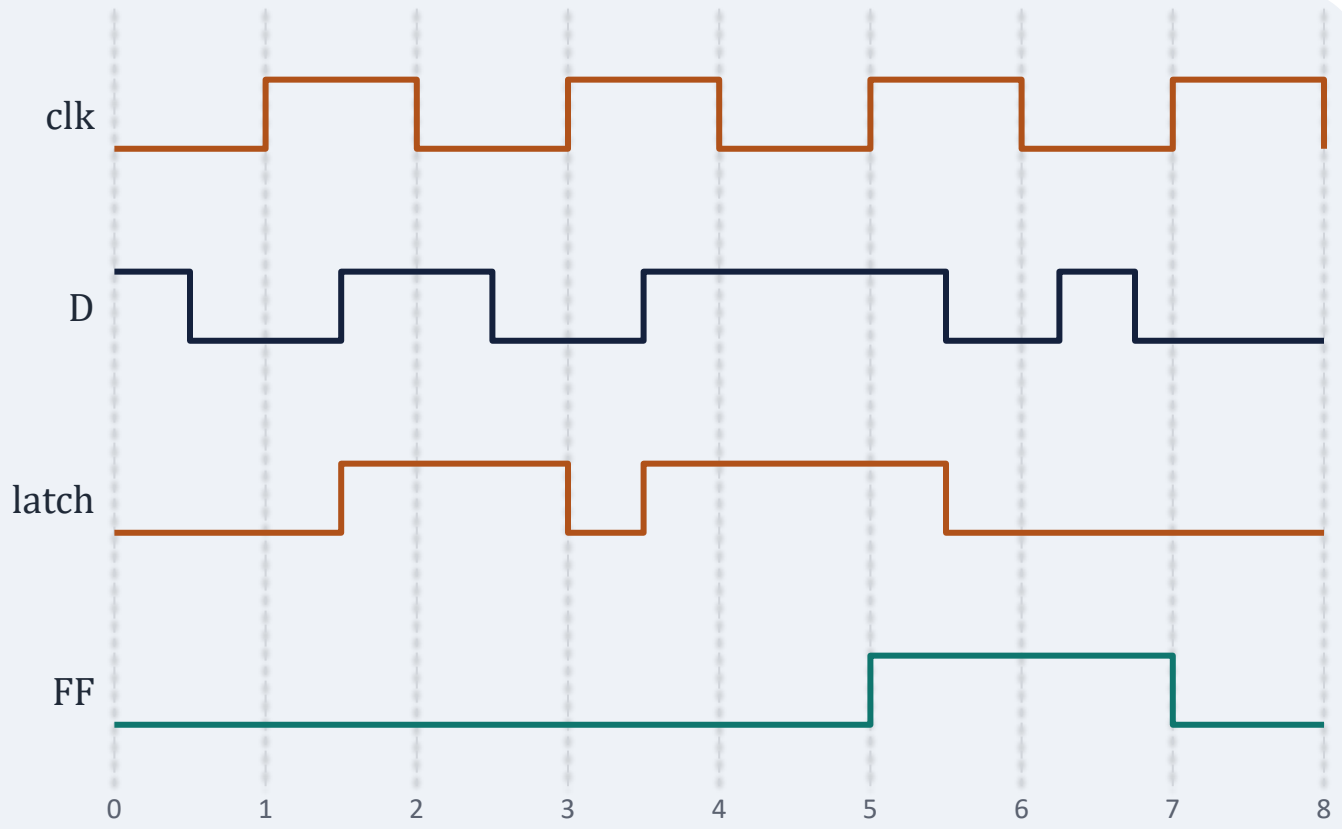
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Review 1 — Latch versus flip-flop

§5.4

REVIEW

Draw Q for a D latch (enable = clk) and a positive-edge D flip-flop. Both start at 0.



1. FF: samples D at rising edges 1, 3, 5, 7
2. Samples 0, 0, 1, 0 → rises at 5, falls at 7
3. Latch: $Q = D$ while clk = 1; holds while clk = 0
4. 1–2: D rises at 1.5 → $Q = 1$
5. 3–4: passes $D = 0$ at 3, then 1 again at 3.5
6. 5–6: D falls at 5.5 → $Q = 0$
7. The 6.25 pulse comes while clk = 0: missed
8. 7–8: $D = 0$ → Q stays 0

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Review 2 — JK trace and conversion

§5.4

REVIEW

A JK flip-flop starts at $Q = 1$. Apply JK = 01, 11, 11, 00, 10. Then make a T flip-flop from a D flip-flop.

Edge	J K	Action	Q
1	0 1	reset	0
2	1 1	toggle	1
3	1 1	toggle	0
4	0 0	hold	0
5	1 0	set	1

1. Rules: 00 hold, 01 reset, 10 set, 11 toggle
2. Start $Q = 1$. Edge 1: 01 → reset: $Q = 0$
3. Edge 2: 11 → toggle: 1; edge 3: 11 → toggle: 0
4. Edge 4: 00 → hold: 0; edge 5: 10 → set: 1
5. Q = **0, 1, 0, 0, 1**
6. T from D: we want $Q^+ = T \oplus Q$; a D gives $Q^+ = D$
7. Equate them: $D = T \oplus Q$, one XOR fed from its own Q

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Review 3 — Maximum clock frequency

§5.4

REVIEW

$t_{pd} = 6$, $t_{setup} = 4$, $t_{hold} = 2$ ns. Longest logic path 18 ns, shortest 1 ns. Find f_{max}; check hold.

The terms

t_{pd}, propagation delay: clock edge until Q shows its new value (clock-to-Q).

t_{comb}: delay through the logic; max = slowest path, min = fastest.

t_{setup}: D stable this long BEFORE the edge.

t_{hold}: D stable this long AFTER the edge.

T: clock period; $f_{max} = 1/T_{min}$.

The two constraints

$T \geq t_{pd} + t_{comb,max} + t_{setup}$

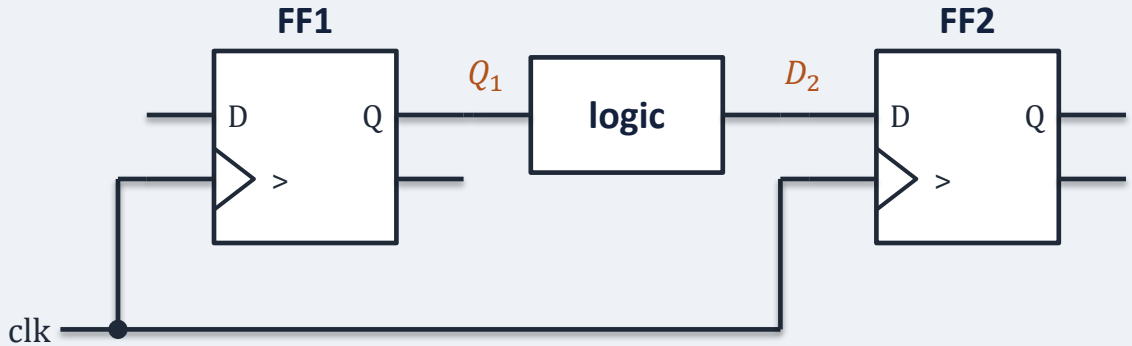
$t_{pd} + t_{comb,min} \geq t_{hold}$

1. Setup: $T \geq t_{pd} + t_{comb,max} + t_{setup}$
2. $= 6 + 18 + 4 = 28$ ns
3. $f_{max} = 1/28$ ns $=$ **35.7 MHz**
4. Hold: earliest change $t_{pd} + t_{comb,min} = 6 + 1 = 7$ ns
5. $7 \geq 2$ ✓; margin $7 - 2 =$ **5 ns**
6. Hold fails only if $t_{pd} + t_{min} < 2$ ns
7. A hold violation cannot be fixed by slowing the clock

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Review 3 — Timing to scale

§5.4



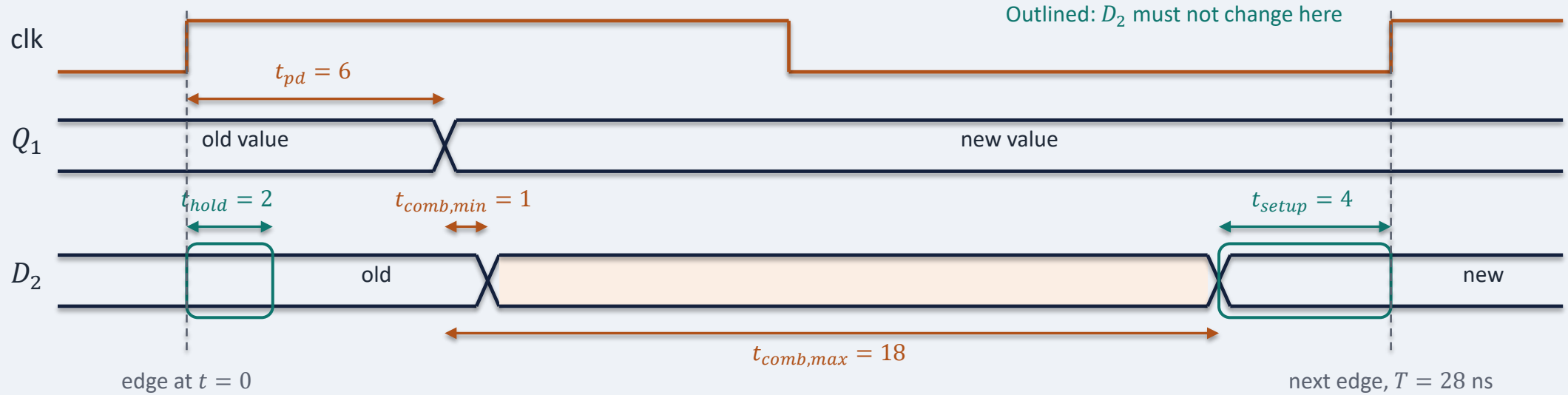
t_{pd}, propagation delay (clock-to-Q): from the clock edge until Q shows its new value

t_{comb}: delay through the logic (min = fastest path, max = slowest)

t_{setup}: D must be stable this long BEFORE the edge

t_{hold}: D must stay stable this long AFTER the edge

Here: setup margin $24 - 24 = 0$ ns; hold margin $7 - 2 = 5$ ns



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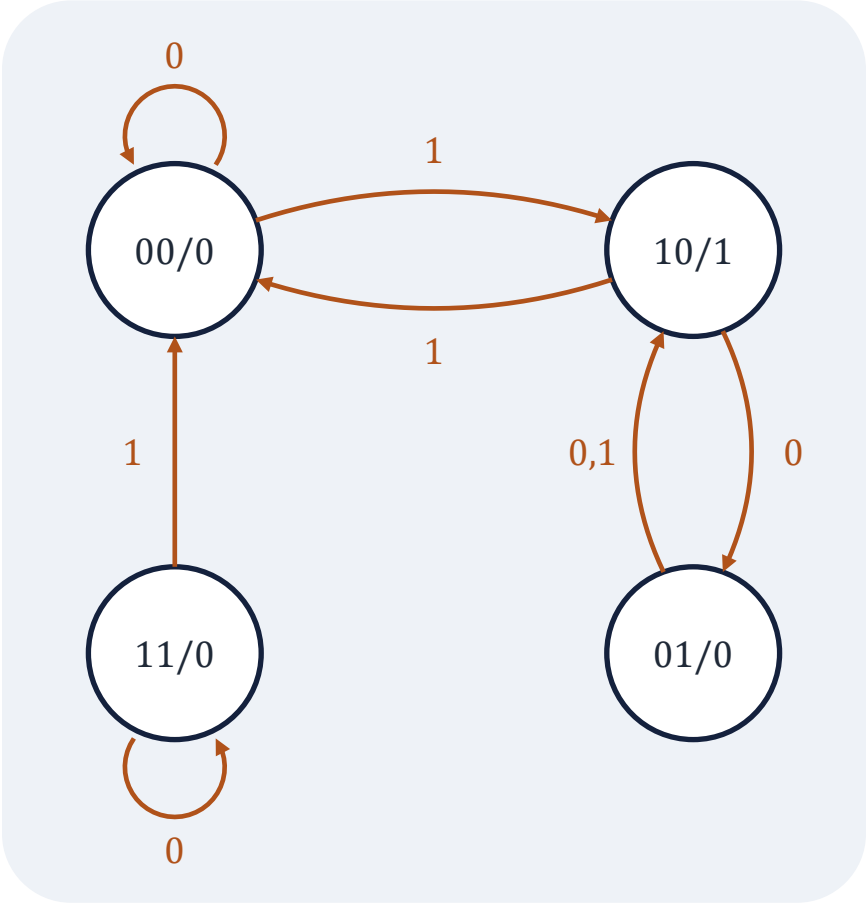
Review 4 — Analyze a circuit

§5.5

REVIEW

$D_A = A'x + Bx'$, $D_B = Ax'$, $y = AB'$. Find the state table and state diagram. Mealy or Moore?

A B	x	A^+ B^+	y
0 0	0	0 0	0
0 0	1	1 0	0
0 1	0	1 0	0
0 1	1	1 0	0
1 0	0	0 1	1
1 0	1	0 0	1
1 1	0	1 1	0
1 1	1	0 0	0



1. D flip-flops: $A^+ = D_A$, $B^+ = D_B$
2. $A^+ = A'x + Bx'$
3. $B^+ = Ax'$
4. Each row: put A, B, x into both
5. e.g. 10, $x = 0$: $A^+ = 0$, $B^+ = 1$ → 01
6. $y = AB'$ uses the state only: **Moore**
7. $y = 1$ only in state 10

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Break

10 minutes — we resume at the 0:55 mark

Next: state reduction, design, registers and counters

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Review 5 — Reduce the states

§5.7

REVIEW

Reduce this state table (next state and output for $x = 0$, $x = 1$).

State	Next x=0	Next x=1	y x=0	y x=1
a	b	c	0	0
b	d	e	0	0
c	f	e	0	0
d	a	e	0	1
e	a	e	0	1
f	a	e	0	1

1. Group by outputs: $\{a, b, c\}$ give 0, 0; $\{d, e, f\}$ give 0, 1
2. d, e, f all go to (a, e): identical → $d \equiv e \equiv f$
3. Replace e and f by d everywhere
4. Now $b \to (d, d)$ and $c \to (d, d)$: identical → $b \equiv c$
5. $a \to (b, b)$ differs from $b \to (d, d)$: a stays alone
6. Classes $\{a\}, \{b, c\}, \{d, e, f\}$: **six states → three**
7. Two flip-flops instead of three

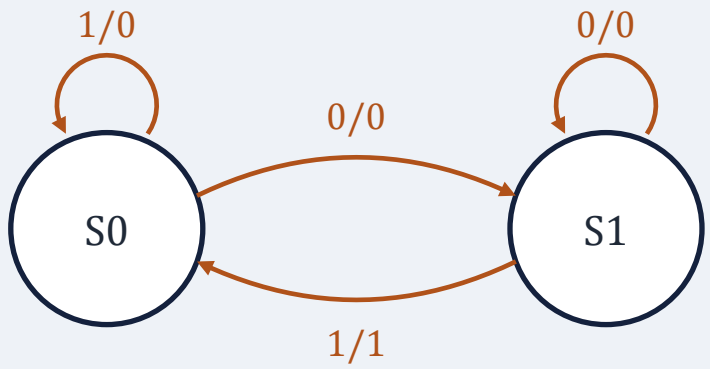
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Review 6 — Design a detector

§5.8

REVIEW

Mealy machine: $y = 1$ when the input sequence ends in 01. Use one D flip-flop.



S1 = ‘the last input was 0’.

A	x	A^+	y
0	0	1	0
0	1	0	0
1	0	1	0
1	1	0	1

1. S_1: the last input was 0
2. Codes: $S_0 = 0$, $S_1 = 1$ (one flip-flop A)
3. From any state: $x = 0 \rightarrow S_1$; $x = 1 \rightarrow S_0$
4. So $A^+ = x'$: $D_A = x'$
5. $y = 1$ in S_1 with $x = 1$ (ends in 01)
6. $$y = Ax$$
7. Mealy: y uses x

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Review 7 — Shift register trace

§6.2

REVIEW

A shift-right register holds 0110. The serial input into the MSB is 1, 1, 0, 1. Trace it.

Clock	SI	$A_3A_2A_1A_0$
0	—	0110
1	1	1011
2	1	1101
3	0	0110
4	1	1011

1. Every clock: bits move right; SI enters A_3; A_0 is lost
2. Clock 1, SI = 1: 0110 → 1011
3. Clock 2, SI = 1: 1011 → 1101
4. Clock 3, SI = 0: 1101 → 0110
5. Clock 4, SI = 1: 0110 → **1011**
6. After 4 clocks it holds the 4 serial bits; the oldest is in A_0

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Review 8 — Counters

§6.4

REVIEW

(a) Build a mod-6 counter from a 4-bit counter with synchronous load. (b) Compare a 4-bit ring and Johnson counter.

(a) Mod 6: 0 … 5

At 5 = 0101, assert Load with inputs 0000.

$Load = Q_2 Q_0$ is enough: among 0–5 only 5 has both bits.

(b) Ring vs Johnson

Ring: 4 states, one-hot, no decoder.

Johnson: 8 states, 2-input AND to decode each.

1. (a) Mod 6 counts 0–5, then back to 0
2. At 5 = 0101, load 0000 on the next edge
3. Load must be 1 only at 5: only 5 has $Q_2 = Q_0 = 1$
4. So $Load = Q_2 Q_0$, a 2-input AND
5. Synchronous load acts at the edge: no glitch
6. (b) Ring feeds back Q_0; Johnson feeds back Q_0'
7. 4 bits: ring **4** states, Johnson **8** states

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Practice set — 15 minutes

§5.8

YOUR TURN

≈ 15 min

1. $t_{pd} = 5$, $t_{comb} = 10$, $t_{setup} = 5$ ns. Maximum clock frequency?
2. A JK flip-flop has $J = K = 1$ and $Q = 0$. What is Q after one edge? After two?
3. Write the excitation table of a T flip-flop.
4. How many flip-flops does a 12-state machine need (binary)?
5. Which counter has no false counts: ripple or synchronous? Why?

Work alone, as on the exam.

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Practice set — Solutions

§5.8

SOLUTION

1. $T \geq t_{pd} + t_{comb} + t_{setup} = 5 + 10 + 5 = 20$ ns

2. $f_{max} = 1/20$ ns $=$ **50 MHz**

3. 2. $J = K = 1$ toggles: $Q = 0 \rightarrow$ **1**; next edge → **0**

4. 3. $T = Q \oplus Q^{+}$: $0 \rightarrow 0$: 0; $0 \rightarrow 1$: 1

5. $1 \rightarrow 0$: 1; $1 \rightarrow 1$: 0

6. 4. n flip-flops give 2^n states: $2^3 = 8 < 12 \leq 16 = 2^4$

7. So **4 flip-flops**

8. 5. **Synchronous**: every bit changes on the same edge

9. Ripple: bits change one after another → false counts

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Key takeaways

D

$$Q^+ = D$$

JK

$$Q^+ = JQ' + K'Q$$

T

$$Q^+ = T \oplus Q$$

Clock

$$T \geq t_{pd} + t_{comb} + t_{su}$$

Design

diagram → table → maps

Counters

check unused states

Sequential design is a procedure: follow it every time.

Homework #11 due at the start of Lecture 28

Next: Exam 3, then random-access memory

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